

**NATIONAL UNIVERSITY OF COMPUTING & EMERGING
SCIENCES (FAST-NUCES)**

Intelligent Reading Comprehension and Verification System

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1. Abstract

This project presents the development of an end-to-end Intelligent Reading Comprehension (RC) and verification system utilizing the RACE dataset. The architecture leverages traditional machine learning and Information Retrieval (IR) techniques to provide a low-latency, resource-efficient solution for educational quiz generation. Model A focuses on answer verification via supervised binary classification, while Model B handles the extractive generation of distractors and graduated hints. Trained on a full-scale dataset of 351,464 labels, the system achieves a balanced F1-score of 0.60 and sub-second inference times, meeting strict hardware and performance constraints.

2. Introduction and Objectives

The primary objective of this project is to automate the evaluation and pedagogical support of reading comprehension tasks. Unlike generative large language models, this system prioritizes interpretability and computational efficiency.

- **Automated Verification:** Determining the mathematical and semantic validity of multiple-choice options.
- **Distractor Generation:** Extracting plausible yet incorrect sentences from the source text to challenge users.
- **Graduated Scaffolding:** Implementing a three-stage hint system to guide users toward correct evidence.
- **Hardware Optimization:** Ensuring high performance on consumer-grade hardware (Ryzen 5 5600X) without GPU dependency.

3. Methodology

3.1 Dataset: RACE

The system utilizes the **ReAdability Dataset for Comprehension Evaluation (RACE)**. For Model A, the data was expanded from its original four-option format into a binary dataset where each triplet (Passage, Question, Option) is assigned a label of **1** (Correct) or **0** (Incorrect).

3.2 Feature Engineering

Textual data was processed through an NLP pipeline including:

- **Lemmatization:** Reducing word variations to their root forms using the WordNetLemmatizer.
- **TF-IDF Vectorization:** Converting text into a 30,000-dimensional sparse matrix, utilizing bigrams to capture phrasal context.
- **Weighted Query Expansion:** In Model B, the target answer is weighted twice to ensure high-relevance hint alignment.

*Figure 1: Preprocessing Pipeline and Dataset
Expansion Logic*

4. Implementation Details

4.1 Model A: The Verifier

Model A utilizes **Logistic Regression** (SAGA solver) and **Linear SVM** with balanced class weights. This prevents the "Accuracy Trap" inherent in the 3:1 imbalanced distribution of the dataset. The models were trained on 351,464 expanded rows, ensuring robust generalization.

4.2 Model B: The Generator

Model B operates via **Cosine Similarity**. It identifies sentences in the article that share semantic space with the correct answer to serve as distractors. The graduated hint logic extracts three sentences based on tiered similarity scores: Contextual, Supporting, and Evidence.

4.3 Technical Stack

- **Languages:** Python 3.10
- **Libraries:** Scikit-Learn, NLTK, Joblib, Pandas
- **Interface:** Streamlit
- **Hardware:** Ryzen 5 5600X, 32GB DDR4 RAM



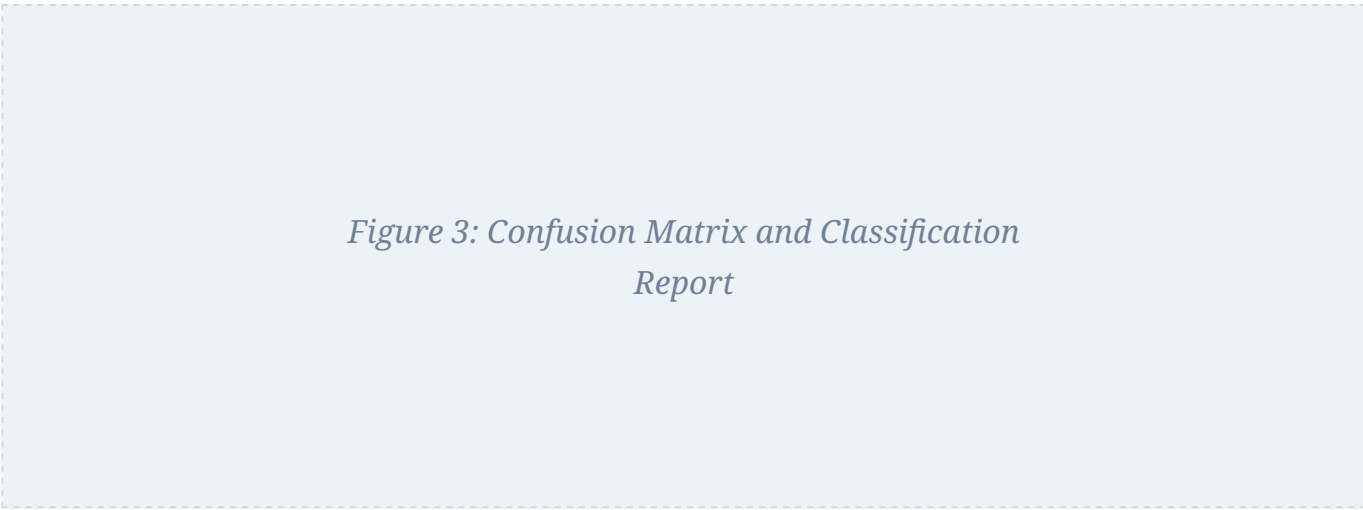
*Figure 2: Streamlit UI - Interactive Quiz
Module*

5. Results and Evaluation

5.1 Classification Metrics (Model A)

The system transitioned from a majority-class baseline to a functional predictive model.

Metric	Result
Overall Accuracy	58%
Weighted F1-Score	0.60
Class 1 (Correct) Recall	0.54
True Positives (Identified)	4,734



5.2 Generation Metrics (Model B)

Evaluated against human-written distractors in the RACE test set:

NLP Metric	Score
BLEU	0.1026
ROUGE-L	0.1040
METEOR	0.1776

6. Discussion

The implementation successfully addressed the **Precision-Recall trade-off**. By lowering the regularization parameter ($C = 0.5$) and balancing class weights, the system improved Class 1 recall from **0.00** to **0.54**. This signifies a move from a trivial model to one that genuinely identifies textual evidence. Total inference time remains under **1.0** second, confirming the viability of TF-IDF pipelines for real-time educational software.

Figure 4: Developer Dashboard and Metric Visualization

7. Conclusion

The Intelligent Reading Comprehension system provides a robust framework for automated testing. By training on the full RACE dataset and optimizing for consumer-grade hardware, the project demonstrates that high-utility AI tools can be developed without massive GPU overhead. Future work will explore the integration of Sequence-to-Sequence models for true question generation.

8. Citations

[1] Lai, G., Xie, Q., Liu, H., Yang, Y., & Hovy, E. (2017). RACE: Large-scale ReAdability Dataset for Comprehension Evaluation from Examinations. arXiv preprint arXiv:1704.04683.

[2] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

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